

Extension Tasks

The following tasks are designed to make the Patient Visit Manager more realistic.

1. Time-Slot Conflict Detection

- Prevent overlapping appointments for the same patient within a 30-minute window. Display a warning and optionally allow override.
- Implementation Guide:
- When adding a new visit, manually loop through the existing visit list.
- For each visit, check:
- If the PatientName matches
- If the VisitDateTime is within ± 30 minutes of the new visit
- If a conflict is detected, print a warning:
- "Warning: This patient has another visit within 30 minutes. Proceed? (Y/N)"
- Read user input and proceed or cancel accordingly.

2. Filtering and Sorting

- Allow filtering and sorting of visit records via menus.
- Implementation Guide:
- Prompt users with filter options:
- Filter by doctor name, type, or date range

3. Role-Based Console Access

- Create roles: Admin (full access) and Receptionist (limited to add/search/report). Prompt for login with hardcoded credentials.
- Implementation Guide:
- Create an enum for UserRole { Admin, Receptionist }
- At program start, ask for role selection (hardcoded username/password).
- Based on role, show or restrict options:
- Admin sees full menu
- Receptionist sees limited menu (e.g., only Add/Search/Report)

4. User Activity Logging

- Maintain a log file that records all user actions with timestamps and outcome status (success/failure).

- Implementation Guide:
- Create or append to activity_log file.

5. Visit Duration and Fee Calculation

- Add visit duration and calculate fees based on type. Load rates from a JSON config file for easy updates.
- Implementation Guide:
- Add DurationInMinutes property to Visit class.
- Store fee rules in a fees.json file:

```
{  
  "Consultation": 500,  
  "Follow-up": 300,  
  "Emergency": 1000  
}
```